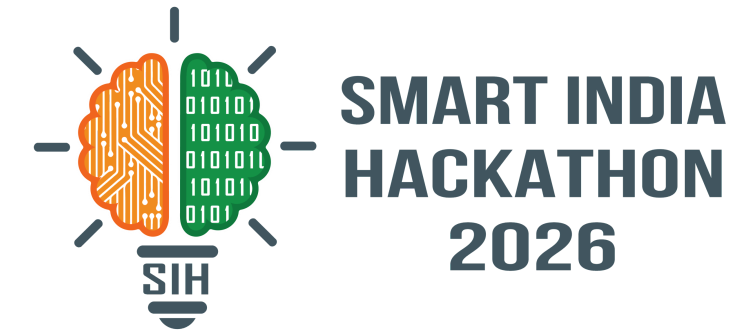
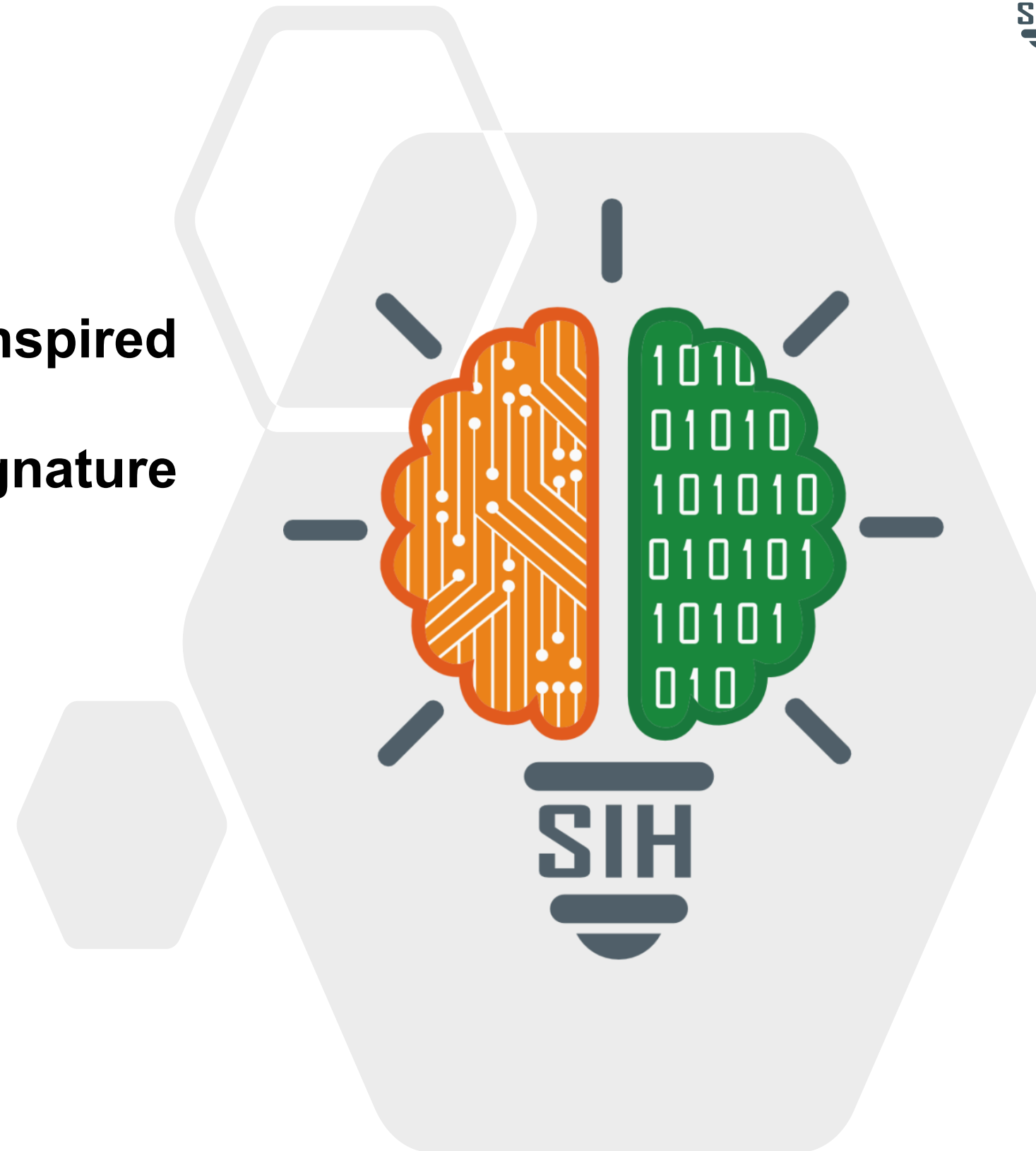


SMART INDIA HACKATHON 2026



- **Problem Statement ID – SIH26141**
- **Problem Statement Title – Quantum-Inspired Cyber Threat Detection for Digital Signature Security**
- **Theme- Blockchain & Cybersecurity**
- **PS Category – Software**
- **Team ID –**
- **Team Name – Order 66**



Problem Overview

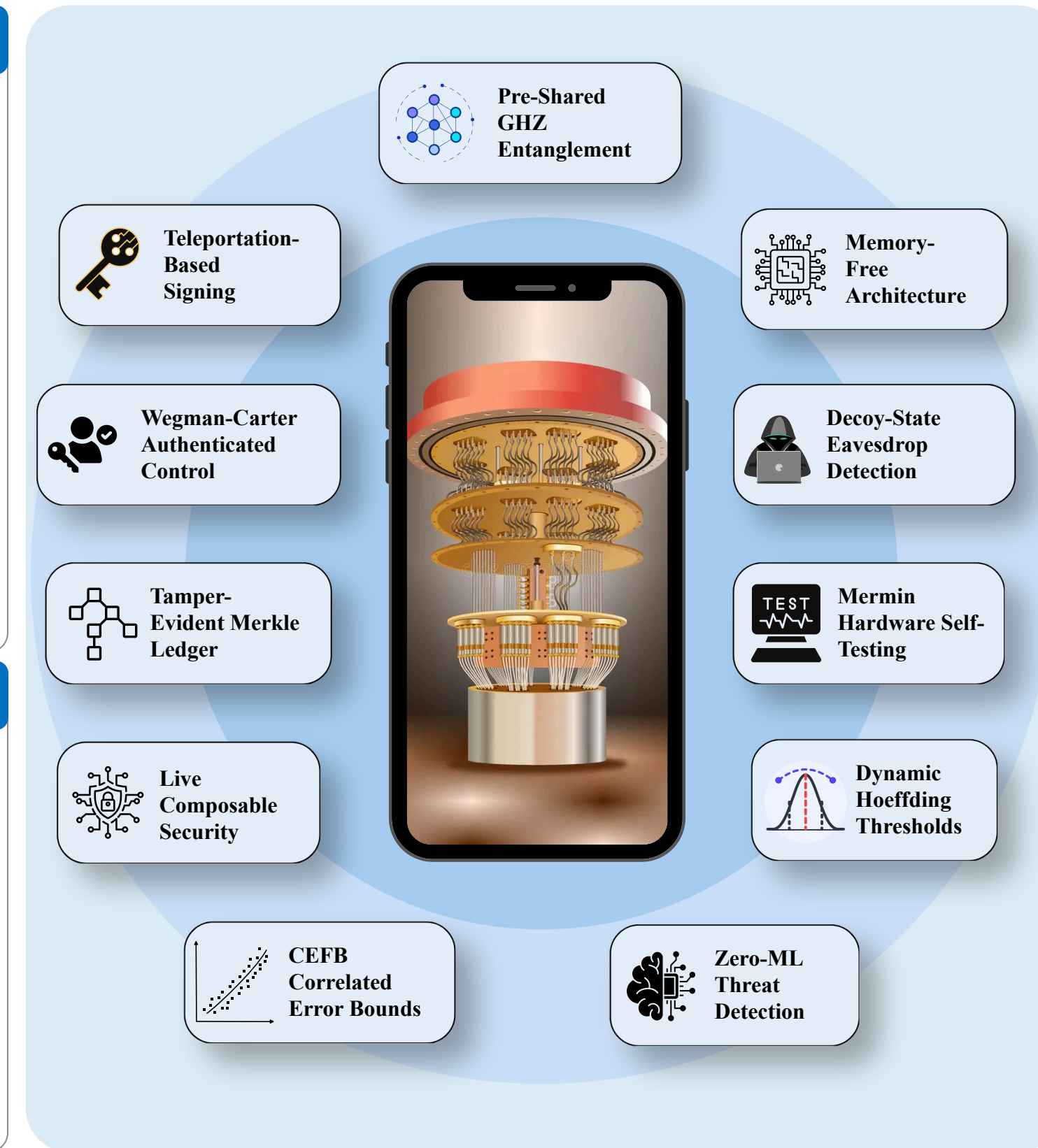
- **Quantum-Vulnerable Signatures:** RSA/ECDSA break under Shor's algorithm — intercepted data is already at risk via "harvest-now-decrypt-later" attacks.
- **Impractical Legacy QDS:** Existing models (e.g., Gottesman-Chuang) need unscalable long-term quantum memory.
- **Transmission Bottleneck:** Live fiber-optic quantum transmission is limited by attenuation and decoherence.
- **Exposed Classical Layer:** Research focuses only on the quantum channel — authentication and logging stay unprotected.

Innovation & Uniqueness

- **Information-Theoretic Security:** Unbreakable by any computing power — grounded in physics, not unproven math.
- **Device-Level Detection:** Mermin self-test catches hardware tampering that statistical checks alone miss.
- **Zero-ML, Fully Explainable:** Every detection decision is deterministic and reproducible.
- **Novel CEFB Bound:** A genuine mathematical advance over standard QKD thresholding.
- **Multi-Layered Defense:** Quantum layer stops interception; Merkle layer secures classical bookkeeping.

Detailed Solution

- **Pre-Shared Entanglement:** GHZ states pre-distributed so signing needs just 2 classical bits/party — no live quantum transmission.
- **Decoy-State Detection:** Exploits the no-cloning theorem; eavesdropping disturbs decoys and raises error rates.
- **Hardware Self-Test:** Mermin/GHZ batches catch detector tampering that breaks real-world QKD.
- **Dual Statistical Bounds:** Hoeffding limits + a martingale-corrected bound (CEFB) guard against correlated errors.
- **Hardened Control Plane:** Wegman-Carter MACs + Merkle-hashed ledger secure the classical channel independently.



Methodology and Process

- **State Modeling:** C++ engine simulates quantum state vectors, Pauli operators, GHZ entanglement with finite-key accuracy.
- **Secured Control Layer:** Authenticated, timing-screened frames block replay, burst, spoofing.
- **Hardware Verification:** Dedicated Mermin-inequality batches catch detector-blinding independent of signal traffic.
- **Adaptive Thresholding:** Real-time Hoeffding + martingale-corrected bounds block correlated-error exploits.
- **Decoy-State Protocol:** Randomly interleaves verification qubits to benchmark noise, catch intercept-resend attacks.
- **Live Dashboard:** Full-stack UI shows QBER, threshold margins, anomaly flags for explainable monitoring.

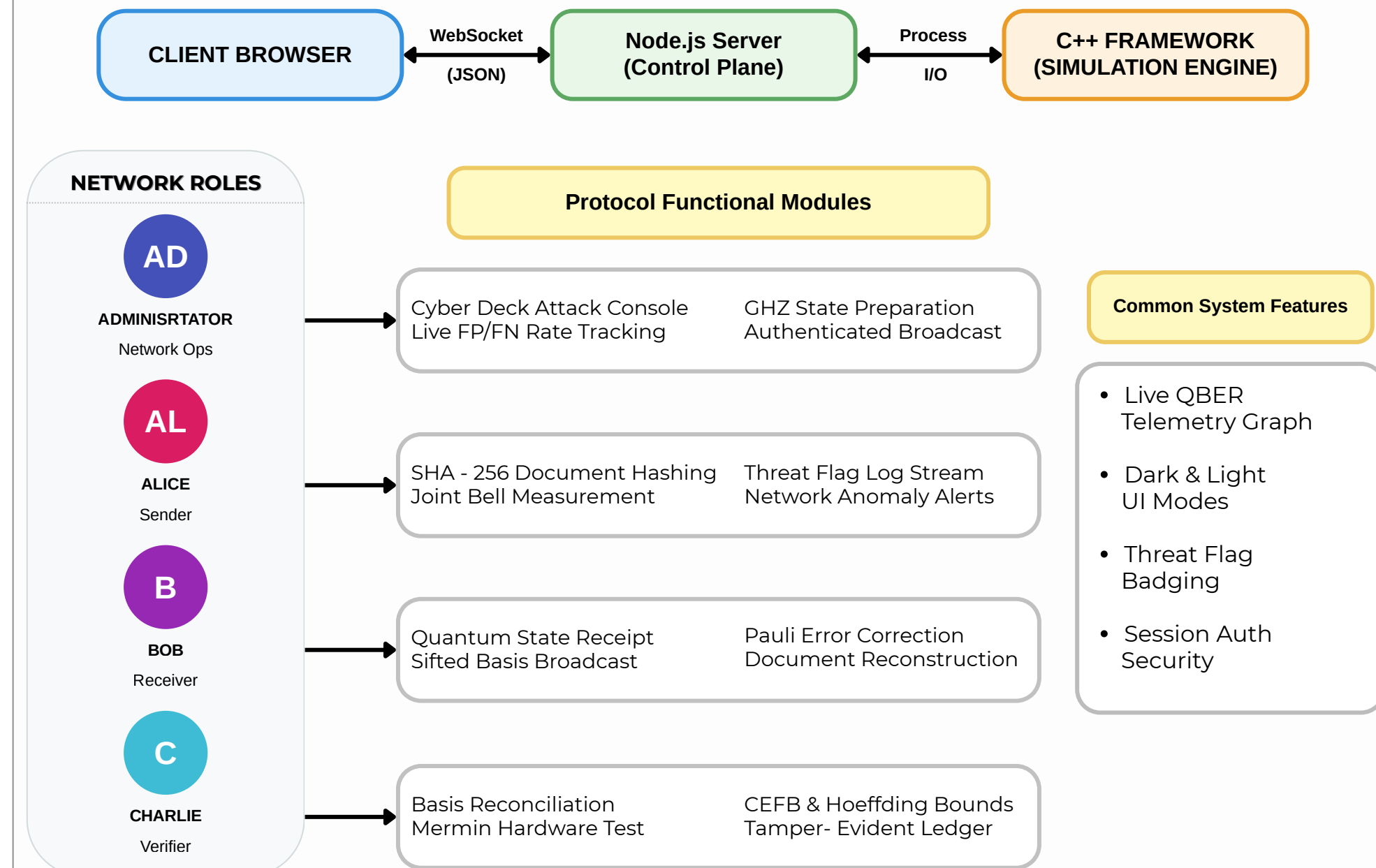
Frontend: React 18, Vite, Tailwind CSS,
Framer Motion, Lucide React

Backend: Node.js, WebSockets, Native HTTP.

Core Simulation: C++17, Eigen3, GCC, MSYS2,
Python

Cryptography: node:crypto,
Wegman-Carter MAC,
Merkle Trees, SHA-256

Version Control: Git, GitHub

**SYSTEM ARCHITECTURE**

Financials



Enterprise Licensing:
Annual subscriptions
for secure corporate
networks.



Defense Contracts:
Custom deployments
for highly classified
government networks.



API / SDK Usage:
Pay-per-signature
model for third-party
financial integrations.



Hardware Integration:
Consulting and
deployment fees with
QKD hardware vendors.



Node.js Control Plane (25):
WebSocket architecture, Wegman-
Carter MACs, threat detection (₹80K
- ₹1L)

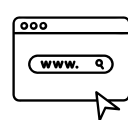
CEFB/Hoe
folding



C++ Physics Engine (35%):
Mathematical bounds
(CEFB/Hoefding), state vector
simulation, optimization (₹1.2L)



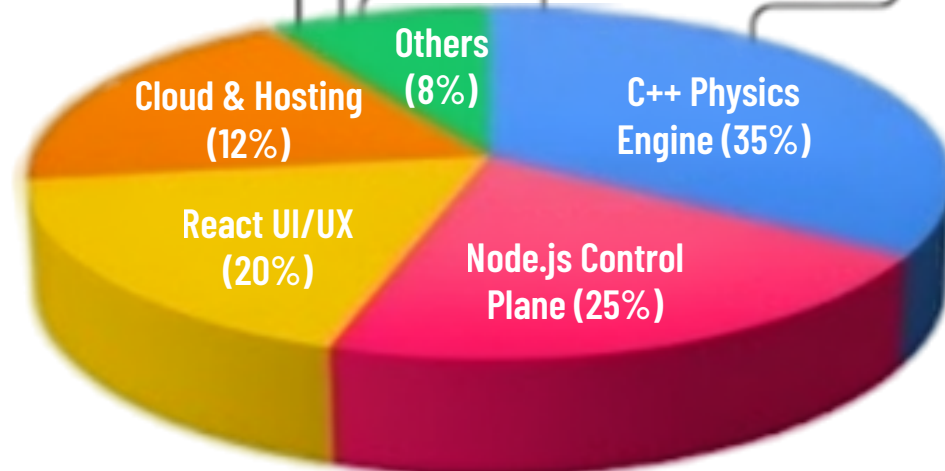
Cloud & Hosting (12%):
Real-time data streaming
servers, high-availability
architecture (₹40K - ₹50K)



React UI/UX (20%):
Interactive dashboard, QBER
telemetry graphs, anomaly
visualization (₹60K - ₹75K)



Others (8%):
Security audits, NIST/ISO
compliance prep, documentation
(₹25K - ₹35K)



SUPPORTING FACTS FOR FEASIBILITY AND VIABILITY



NIST PQC Mandate (2024): Legacy RSA/ECDSA must be phased out now — "harvest-now-decrypt-later" puts intercepted data at risk today.



National Quantum Mission (2023): ₹6,000 Cr committed to secure quantum comms over 2,000 km — proves domestic backing.



Market Growth: Global quantum cryptography market projected at \$3.2B by 2030.

Viability



Harvest-Now Threat: Shor's algorithm creates urgent demand for quantum-safe solutions today.



Regulatory Shift: NIST PQC standardization drives mandatory upgrades across govt/banking by 2030.

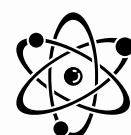


Hardware Agnostic: Designed to integrate with commercial QKD hardware (Toshiba, ID Quantique).

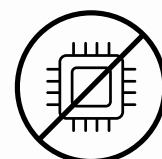


Market Appeal: Merkle auditing opens legal-tech, finance, and e-voting markets.

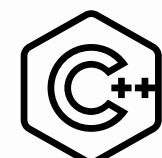
Feasibility



Proven Physics: Relies on established GHZ teleportation and Pauli measurement.



No Quantum Memory Required: Removes the biggest current QDS engineering hurdle.



Simulated, Not Hardware-Bound:
C++/Eigen3 proves the framework without costly quantum hardware.



Deterministic Control Layer: Lightweight Node.js logic — fast, stable, predictable.

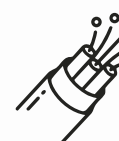
Challenges and Mitigation



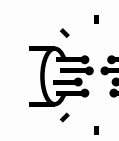
Quantum hardware costs



C++ engine validates bounds pre-deployment.



Correlated fiber noise



Novel CEFB bound guarantees security.



Classical-layer attacks



Wegman-Carter MACs + Merkle ledgers.



Complex telemetry



Visual dashboard with color-coded threat badging.

IMPACT AND BENEFITS

BENEFITS

1. **Information-Theoretic Security:** The platform uses physical quantum laws to provide unconditional security, ensuring signatures remain unbreakable even against future quantum computers.
2. **Memory-Free Architecture:** Utilizing GHZ teleportation allows for immediate Pauli measurements, completely eliminating the need for expensive, unscalable quantum memory.
3. **Hardware-Level Auditing:** Parallel Mermin-inequality self-tests ensure cryptographic verification of physical detectors, catching device tampering that standard channel checks miss.
4. **Cross-Layer Threat Defense:** Uniquely combines physical-layer detection with a hardened classical control plane, proactively thwarting timing anomalies, replays, and spoofing.
5. **Real-Time Cyber Telemetry:** Interactive dashboards provide network operators with live visualizations of Quantum Error Rates (QBER), statistical margins, and dynamic threat flags.
6. **Tamper-Evident Accountability:** A Merkle-hashed log architecture ensures transparent, blockchain-style verification of all administrative access and digital signing events.

National Cyber Sovereignty:
Protects infrastructure and defense networks from harvest-now-decrypt-later breaches.

Financial Resilience:
Secures banking and trading systems against future quantum breaks.

IMPACTS

Commercial Viability:
Removes the biggest deployment barrier - quantum memory.

Global Leadership: Positions India as a pioneer in memory free quantum cryptography.

Research Papers

Teleportation-Based Quantum Signatures

- [Wu et al. \(Entropy, 2022\)](#);
- [Wang & Lu et al. \(J. Applied Physics, 2024\)](#);
- [ScienceDirect \(Results in Physics, 2023\)](#).

Hardware & Implementation Attacks

- [Lydersen et al., Nature Photonics 4 \(2010\)](#).

Cryptographic & Statistical Foundations

- [Mermin, Phys. Rev. Lett. 65 \(1990\)](#);
- [Wegman & Carter, J. Comput. Syst. Sci. 22 \(1981\)](#).

Government Applications

- **National Quantum Mission:** ₹6,003.65 Cr approved (2023-31); satellite-based QKD via ISRO.
- **NQM Priority Fit:** [Cryptography/comms security is an explicit live national R&D goal.](#)

Industry Outlook

- \$2.84B PQC market by 2030, 46.2% CAGR (MarketsAndMarkets, 2025).
- "Harvest-now-decrypt-later" — the industry's own framing for today's captured-data risk.
- 52% of 2024 PQC revenue is lattice-based (Kyber/Dilithium) — shows why a physics-based alternative differentiates.

Our Prototype : <https://github.com/AryanRajendraDalvi/PARADOX>

FEATURE	PARADOX	Gottesman -Chuang QDS	RSA / ECDSA
Quantum-computer resistant (Shor-proof)	✓	✓	✗
Information-theoretic (unconditional) security	✓	✓	✗
Built-in channel eavesdropping detection	✓	✓(Static Threshold)	✗
Adaptive statistical thresholding (tightens with sample size)	✓	✗	N/A
Hardware/detector-tampering detection	✓	✗	N/A
No quantum memory required at verifier	✓	✗	N/A
Deterministic replay-attack detection	✓	⚠ (Partial)	✗
Info-theoretic classical-channel authentication (MAC)	✓	N/A	✗
Tamper-evident audit ledger (blockchain-anchored)	✓	✗	✗
Async / offline-verifier signing support	✓	✗	N/A

Research Workflow

Define Problem Statement



Requirement Analysis



Study Existing Systems



Design Proposed System



Feedback and Refinement